Group 101

1.Cumar Abdiqadir Hassan : administrator

2. Nour Hassan Nour : employee

Somaliland Innovation Zone Student Registration System

- **Project Introduction**

1. What is the name of your project?

The name of my project is **Somaliland Innovation Zone Student Registration System**.

2. Why did you select this project?

I selected this project because many training centers and innovation hubs still manage student applications manually. This system automates the registration and application process.

3. What problem does your project solve?

The system solves problems related to manual student registration, application tracking, communication delays, and record management.

4. Who are the target users of your system?

The target users are:

- Students
- Administrators
- Training Coordinators

5. What are the main objectives of the project?

- Automate student registration.
- Manage training programs efficiently.
- Track applications digitally.
- Improve communication between students and administrators.
- Store data securely.

- **Project Requirements**

6. What are the functional requirements of your system?

- Student Registration
- Student Login

- Program Viewing
- Program Application
- Application Tracking
- Admin Dashboard
- Program Management
- Student Management
- Notifications

7. What are the non-functional requirements of your system?

- Security
- Reliability
- Performance
- Scalability
- User-Friendly Interface

8. How did you gather the requirements for the project?

I gathered requirements by analyzing how training centers manage applications and by researching similar online registration systems.

9. What challenges did you identify before development?

- User authentication
- File uploads
- Database design
- API integration
- Data security

- **Development Tools and Technologies**

10. What programming languages did you use?

- Dart

- PHP
- SQL
- JavaScript

11. What technologies did you use to develop the project?

- Flutter
- PHP API
- MySQL
- XAMPP

12. Which front-end language/framework did you use?

Flutter Framework using Dart.

13. Which back-end language/framework did you use?

PHP REST API.

14. Which database did you use?

MySQL Database.

15. Why did you choose this database?

Because MySQL is reliable, secure, fast, and easy to integrate with PHP.

16. Which code editor or IDE did you use?

- Visual Studio Code
- Android Studio

- **System Design**

17. Did you create an ERD (Entity Relationship Diagram)?

Yes, I created an ERD before development.

18. How many tables are in your database?

The database contains 6 tables.

19. What is the purpose of each table?

Students

Stores student information.

Admins

Stores administrator information.

Programs

Stores training and course information.

Applications

Stores student applications.

Notifications

Stores messages and notifications.

Settings

Stores system configurations.

20. What relationships exist between the tables?

- One Student can have many Applications.
- One Program can receive many Applications.
- One Admin can create many Programs.

21. Did you create a flowchart for the system?

Yes, a flowchart was created to visualize the application process.

22. What design methodology did you follow?

I followed the Agile Development Methodology.

- **Project Features**

23. What are the main features of your system?

- Student Registration
- Login System
- Program Management

- Application Management
- Notifications
- Dashboard
- Reports

24. How does user registration work?

Students fill out a registration form and create an account. The information is stored in the database.

25. How does login and authentication work?

The system verifies the student's email and password before granting access.

26. What reports can the system generate?

- Total Students Report
- Applications Report
- Program Report
- Accepted Students Report

27. What security measures did you implement?

- Password Hashing
- Authentication Tokens
- Input Validation
- SQL Injection Prevention

- **Database Questions**

28. What is a primary key?

A primary key uniquely identifies each record in a table.

Example:
Student_ID

29. What is a foreign key?

A foreign key links one table to another.

Example:
Program_ID in Applications table.

30. Why are relationships important in databases?

They reduce redundancy and improve data integrity.

31. How do you insert data into your database?

Example:

INSERT INTO students(name,email)

VALUES('Ahmed','ahmed@gmail.com');

32. How do you update and delete records?

Update:

UPDATE students

SET name='Ali'

WHERE id=1;

Delete:

DELETE FROM students

WHERE id=1;

33. What SQL queries did you use most frequently?

- SELECT
- INSERT
- UPDATE
- DELETE
- JOIN

- **Testing and Evaluation**

34. How did you test your system?

Through manual testing and functionality testing.

35. What types of errors did you encounter?

- API errors
- Database connection errors
- Validation errors

36. How did you fix those errors?

Using debugging techniques and error logs.

37. What improvements would you make in the future?

- Mobile notifications
- Online interviews
- AI Assistant
- Analytics Dashboard

- **Project Demonstration**

38. Show the login page.

The login page allows students and admins to access the system securely.

39. Show the dashboard.

The dashboard displays system statistics and user activities.

40. Demonstrate adding a new record.

Admin creates a new training program.

41. Demonstrate editing a record.

Admin updates training details.

42. Demonstrate deleting a record.

Admin removes outdated programs.

43. Show a generated report.

Display total registrations and application reports.

- **Future Development**

44. What new features would you add in the future?

- AI Chatbot
- SMS Notifications
- Certificate Generator
- Interview Scheduling

45. How can the project be improved?

By improving security, performance, and scalability.

46. Can the system be used on mobile devices?

Yes. The student application is built using Flutter and supports Android and iOS devices.

47. How would you scale the system for a large organization?

- Cloud Hosting
- Load Balancing
- Dedicated Database Server
- Microservices Architecture

- **Conclusion**

48. What did you learn from this project?

I learned Flutter development, API integration, database management, authentication, and full-stack system development.

49. What was the most difficult part of development?

Developing secure authentication and integrating Flutter with PHP APIs.

50. What was the most enjoyable part of the project?

Designing the user interface and seeing the entire system work successfully.

- **Project Summary**

Student Interface (Flutter Mobile App)

- Register Account
- Login
- View Trainings
- Apply for Programs
- Track Application Status
- Receive Notifications
- Manage Profile

Admin Interface (Website Dashboard)

- Manage Students
- Create Trainings
- Edit Trainings
- Delete Trainings
- Review Applications
- Approve Students
- Reject Students
- Generate Reports
- Send Notifications

Tools Used

- Flutter
- Dart
- PHP API

- MySQL
- XAMPP
- Visual Studio Code

Thank You.